Human-in-the-Loop Artificial Intelligence System for Systematic Literature Review: Methods and Validations for the AutoLit Review Software

Kevin M. Kallmes JD/MA, Jade Thurnham BA,
Marius Sauca MS, Ranita Tarchand MS, Keith R. Kallmes BS, Karl J. Holub BS;
Nested Knowledge, St. Paul, MN, USA.

Introduction

While artificial intelligence (AI) tools have been utilized for individual stages within the systematic literature review (SLR) process, no tool has previously been shown to support each critical SLR step. In addition, the need for expert oversight has been recognized to ensure the quality of SLR findings. Here, we describe a complete methodology for utilizing AI SLR tools with human-in-the-loop curation, as well as validations, time savings, and approaches to ensure compliance with best review practices.

Methods

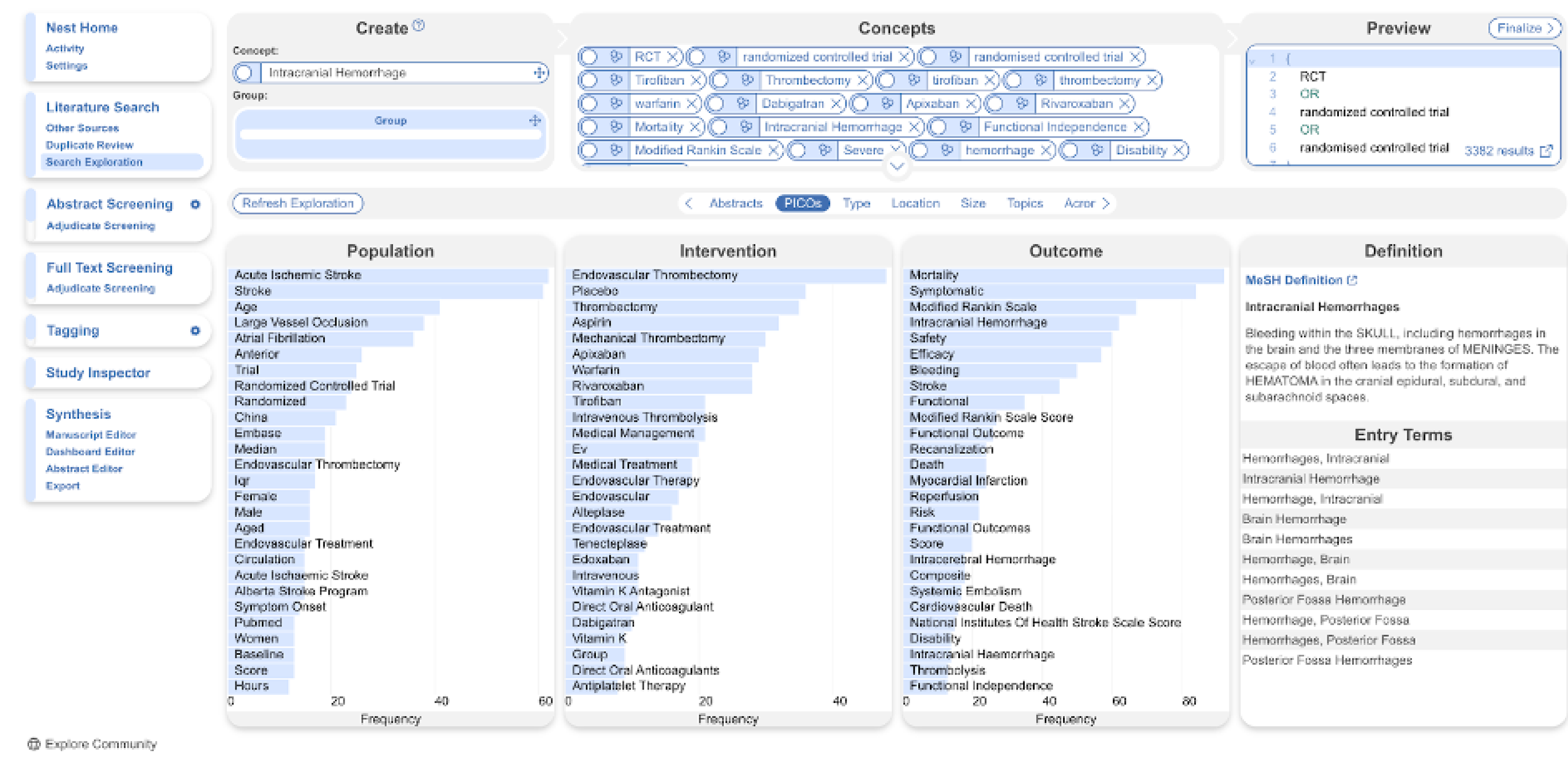
SLRs require completing Search, Screening, and Extraction from relevant studies, with meta-analysis and critical appraisal as relevant. We present a full methodological framework for completing SLRs utilizing AutoLit (Nested Knowledge). This system integrates AI models into the central steps in SLR: Search strategy generation, Dual Screening of Titles/Abstracts and Full Texts, and Extraction of qualitative and quantitative evidence. The system also offers manual Critical Appraisal and Insight drafting and fully-automated Network Meta-analysis. Validations comparing AI performance to experts are reported, and where relevant, time savings and 'rapid review' alternatives to the SLR workflow.

Results

Search strategy generation with the Smart Search AI can turn a Research Question into full Boolean strings with 76.8-79.6% Recall. Supervised machine learning tools can achieve 82%-97% Recall in reviewer-level Screening. Population, Interventions/Comparators, and Outcomes (PICOs) extraction achieved F1 of 0.74; accuracy for study type, location, and size were 74%, 78%, and 91%, respectively. Time savings of 50% in Abstract Screening and 70-80% in qualitative extraction were reported. Extraction of user-specified qualitative and quantitative tags and data elements remains exploratory and requires human curation for SLRs.

Conclusions

AI systems can support high-quality, human-in-the-loop execution of key SLR stages. Transparency, replicability, and expert oversight are central to the use of AI SLR tools.

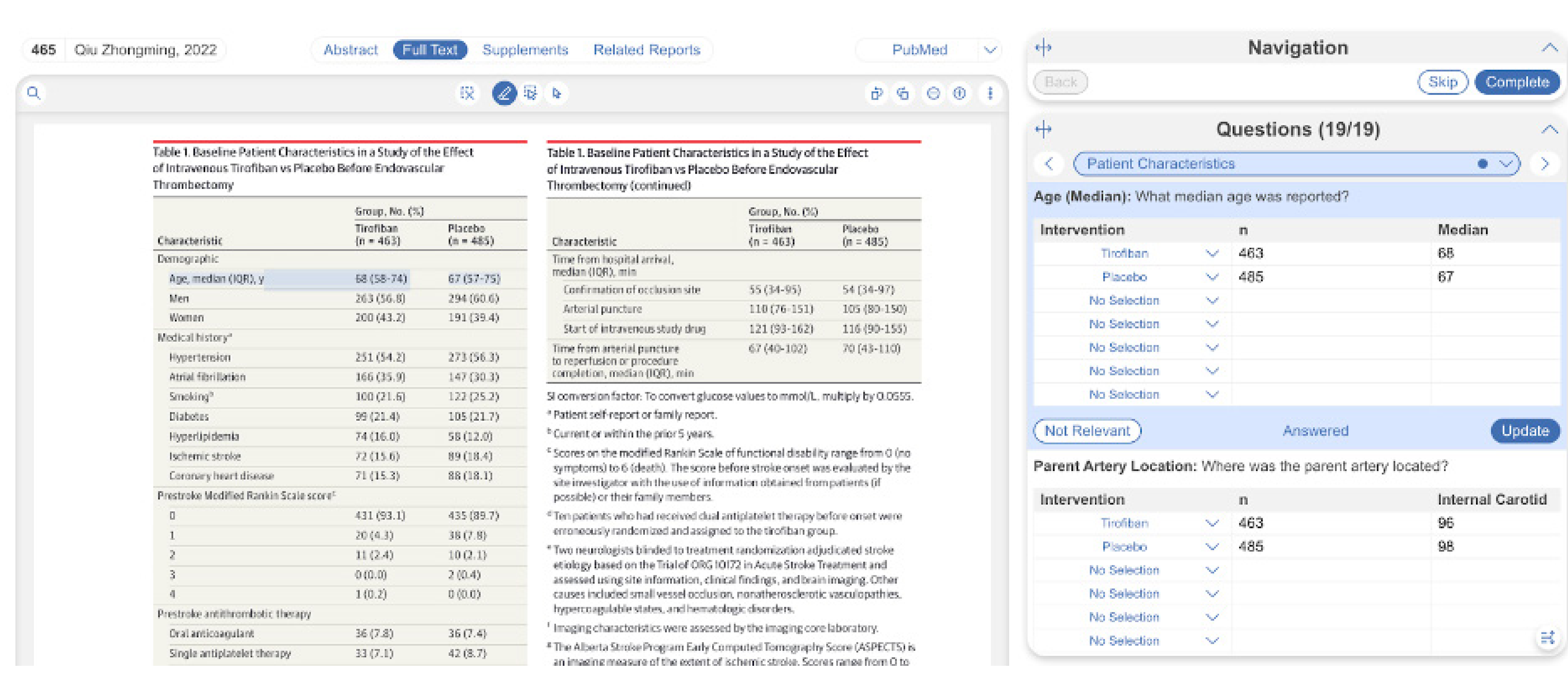


Search Exploration predicts and maps Populations, Interventions, Comparators, and Outcomes from underlying abstracts resulting from Boolean PubMed queries, including reporting frequency and enabling adoption of new search terms.

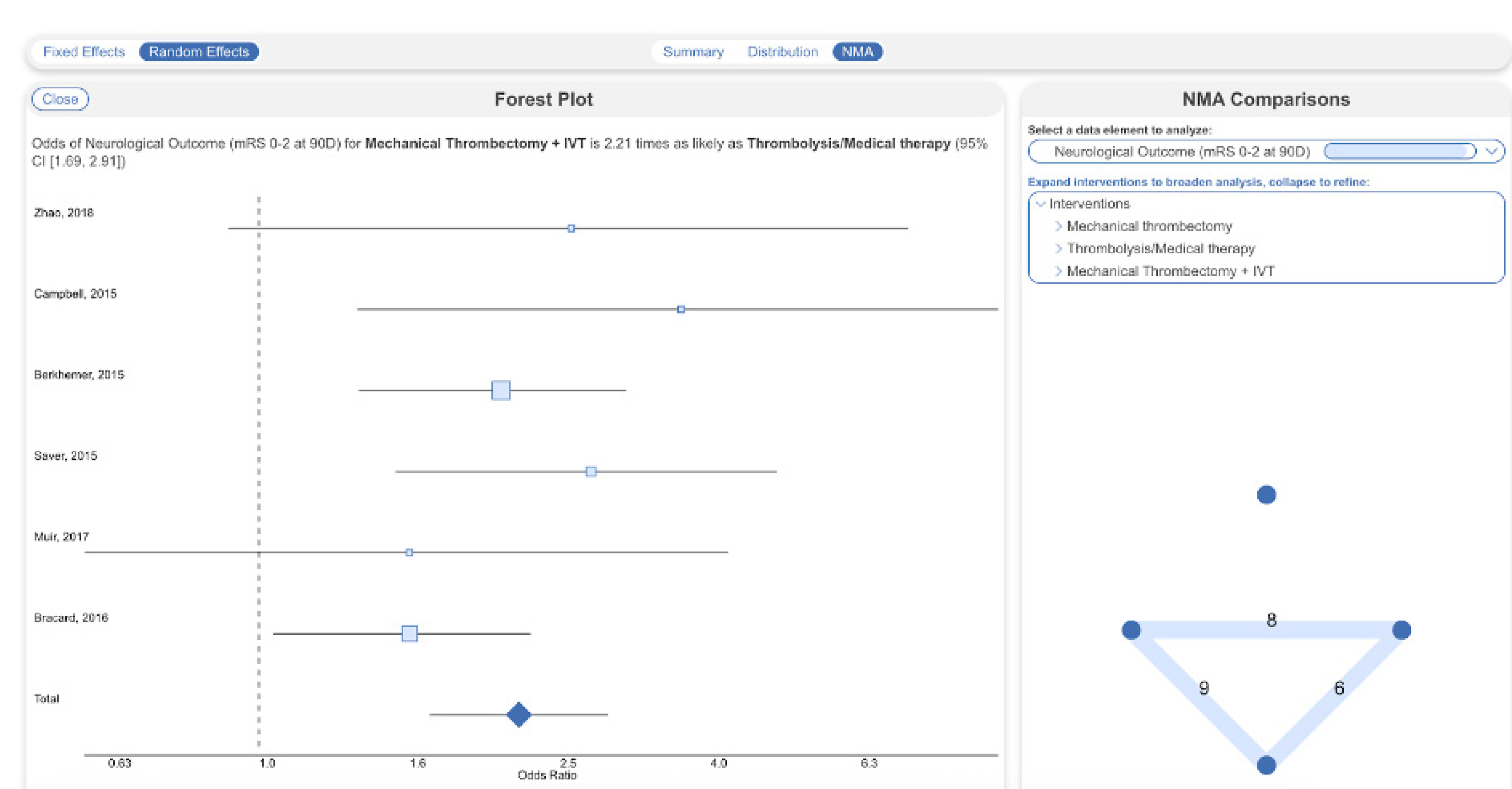
Predictions | Cross Validation

History	Records (adv)	Recall	Precision	F1	Accuracy	AUC
2025-04-07	243 (60)	0.94	0.61	0.74	0.83	0.97
2025-04-01	243 (60)	0.94	0.62	0.75	0.84	0.94
2025-03-20	225 (60)	0.95	0.60	0.72	0.81	0.96
2025-03-20	202 (50)	0.96	0.58	0.72	0.82	0.95
2025-03-20	146 (39)	1.00	0.62	0.76	0.84	0.95
2025-03-19	127 (32)	0.96	0.58	0.73	0.82	0.96
2025-03-19	102 (17)	0.92	0.53	0.67	0.85	0.93
2025-03-19	52 (16)	1.00	0.56	0.72	0.77	0.89

The Inclusion Prediction Model learns from expert screening on a project-by-project basis, with transparent fivefold cross-validation statistics for re-searchers to review regularly.



Adaptive Smart Tags extracts evidence from abstracts or Full Texts (pictured here) based on custom, project-specific Questions. Contents extracted are highlighted for full traceability, and can be text, numeric, options/drop-down, or tabular data (pictured here).PubMed queries, including reporting frequency and enabling adoption of new search terms.



Forest Plot from automated NMA of data extracted in the Meta-analytical Module in AutoLit. Forest plots, Funnel plots, I-squared calculations, SUCRA Rankings, and Odds Ratios with 95% Confidence Intervals, as well as the Network Diagram for hierarchical meta-analysis, are automatically gener-ated in the Quantitative Synthesis module.

References:

Twaites J, Kallmes KM, Holub KJ. Assessment of reasoning agents for building literature search strategies. Value Health. 2025;28(Suppl 1):P22.

Thurnham J, Kallmes KM, Holub KJ. Assessing recall in abstract screening: artificial intelligence vs. human reviewers. Value Health. 2024;27(6):S277.

Twaites J, Kallmes K, Holub K. Repeatable auto-extraction frameworks in clinical systematic literature review: validating a multi-model human-in-the-loop artificial intelligence system for extracting study PICOs, location, size, and type. Value Health. 2025;28(Suppl 1):SA72.

Grys M, Casciano R, Pieniazek I. Comparison of AI-enhanced tools for automating scientific literature reviews. Value Health. 2025;28(Suppl 1):MSR56.

Cichewicz A, Pande A, Borkowska K, et al. Automating systematic literature review (SLR) updates: a comparative validation study of artificial intelligence (AI) versus human screeners. Value Health. 2024;27(6):S263.

Ciapponi A, Bardach A, Glujovsky D, Tarchand R. The efficiency of "Nested Knowledge" to facilitate the conduction of systematic reviews and meta-analyses. Poster presented at: Cochrane Global Evidence Summit; September 4–6, 2024; Prague, Czech Republic. Available from: https://abstracts.cochrane.org/2024-prague-global-evidence-summit/efficiency-nested-knowledge-facilitate-conduction-systematic.

NESTED KNOWLEDGE